Practical Organic Chemistry

Tests of Alcohol :

1) Ceric Ammonium Nitrate (CAN) Test :→

$(NH_4)_2[Ce(NO_3)_6] \xrightarrow{3ROH} \downarrow [Ce(NO_3)_4(ROH)_3] + 2NH_4NO_3$ (Red colour sol^n)

* This Test is given by All Alcohols.
* Amino Acids containing Alcohol → Serine (1° ROH), Threonine (2° ROH) — Also gives CAN Test

2) Victor Meyer Test :→

* Also called [R B C Test] for Alcohols. (Red, Blue, colourless)

Where,
- 1° ROH → Gives Red Colour
- 2° ROH → Gives Blue Colour
- 3° ROH → Gives No Colour (No Rx^n)

1) (i) PCl_3 (ii) $AgNO_2$ (iii) $NaNO_3/HCl$ (iv) Aq. Base → (Red Coloured)

2) (i) PCl_3 (ii) $AgNO_2$ (iii) $NaNO_3/HCl$ (iv) Aq. Base → (Pseudo Nitrol) (No Tautomerisation)

3) → No Rx^n (Colourless Sol^n)

Test of Phenol :

1) Neutral $FeCl_3$ Test :→

* This test is given by Phenol derivatives and enols. (Basically species in which O^- shows Resonance).
* Violet colour observed.

$+ FeCl_3 \longrightarrow Fe(\text{Ar}O^-)_3$ (Violet)

2) Liebermann Nitroso Test :→

* Also called 'GRB Test' (Green-Red-Blue).
* In this test, first Green/Blue colour is observed, then red colour obtained upon dilution and finally blue colour reappears upon adding NaOH.

$\xrightarrow[NaNO_2/HCl]{HNO_2}$ → Indophenol (Green/Blue) $\xrightarrow[\text{Dilution}]{H_2O}$ Red Colour $\xrightarrow[(\text{Alkali})]{NaOH}$ Blue Colour Reappear

Test for Biomolecules :

- Carbohydrates
  1) Molisch Test (Carbohydrate)
  2) Barfoed's Test (Mono & DP Saccharide)
  3) Selwanoff Test (Aldose & Ketose)
  4) Picric acid test (Aromatic proteins)
- Amino Acids
  1) Ninhydrin Test (AAs)
  2) Biuret Test (Protein)
  3) Xanthoproteic Test
  4) Millon's / Cole Test (Phenolic AA)

(I) Test for Carbohydrates :

1) Molisch Test :→

* Shown by all carbohydrates.
* Appearance of a 'Purple Ring' at junction of test sol^n & molisch reagent (α-Naphthol dissolved in ethanol).
* Carbohydrate is converted into furfural form and then it react with α-Naphthol by EAS mechanism.

D-Glucose $\xrightarrow[\text{conc. HCl}]{H_2SO_4}$ (Furfural) $\xrightarrow{\text{EAS}}$ $\xrightarrow{-H^+}$ (Purple Colour Ring)

2) Barfoed Test :→

* Used to detect reducing monosaccharides.
* Disaccharides also react, but very slow, thus can be distinguished from mono.

Reagent : $Cu(OCOCH_3)_2$ / CH_3COOH

* It gives 'Red ppt' in slightly acidic medium

D-Glucose $+ 2Cu^{2+} + H_2O \xrightarrow{(pH<7)}$ D-Gluconic acid $+ Cu_2O\downarrow$ (Red)

3) Selwanoff's Test :→

* Used to detect Ketose carbohydrate.
* Aldose also react, but very less.

Reagent ⇒ Resorcinol

* It gives 'Red ppt' with Ketose

(Furfural form) $\xrightarrow{\text{EAS}}$ $\xrightarrow{-H^+}$ (Red ppt)

NOTE : Mono Ketose ⇒ Deep cherry Red
Di-Ketose ⇒ cherry Red only

4) Picric Acid Test :→

* Very sensitive test, used for detection of 'Reducing Sugars'.

Reagent : Picric Acid

* It gives 'Mahogany Red colour' with reducing sugars.

(Picramic Acid) (Mahogany Red)

NOTE:
- Fehling Test → In basic medium → Distinguish aldehyde & ketone → Gives red colour due to $Cu^{2+} \rightarrow Cu^{+1}$
- Barfoed Test → In Acidic medium → Distinguish mono & disaccharides → Gives red colour due to $Cu^{2+} \rightarrow Cu^{+1}$

(II) Test for Amino Acids :

1) Ninhydrin Test :→

Ninhydrin Preparatory Rx^n

* Used to detect presence of
  1) Ammonia
  2) 1° amine
  3) 2° amine
  4) Amino Acid.

Reagent : Ninhydrin / 2,2-dihydroxy indane-1,3-dione.

* White crystalline solid
* Strong OA
* Highly stable due to Intra H-bond.

* Ninhydrin gives coloured compounds with amino acids as.

AA with 1° amine ⇒ Ruhemann's Purple
AA with 2° amine (Proline) ⇒ Yellow to Orange Range
AA with amide (Asparagine) ⇒ Brown

1) Amino acid 2) Δ 3) H_3O^+ 4) Ninhydrin → (Purple Complex)

→ (Yellow Complex)

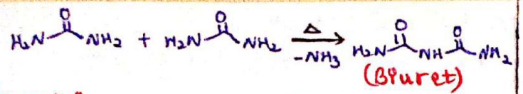
NOTE: Ninhydrin test used to detect finger prints. It reacts with AAs of sweat on fingers.

2) Biuret Test :→

* Generally used to detect Proteins having peptide linkages.
* It can also be used to detect compounds containing atleast 2 of following groups.

$-CH_2-NH_2$, $-\overset{R}{C}-NH_2$, $-\overset{R}{C}-NH-$, $-CS-NH-$, $-\overset{NH}{C}-NH_2$

* Biuret is nowhere used in this test, but since biuret also gives this test, hence named on it.



Reagents:

(i) $CuSO_4 \cdot 5H_2O$

(ii) Sodium Potassium Tartarate [Chelate & stabilise Cu^{2+} salt]

(iii) NaOH [Make med. basic]

(iv) KI [Resist reduction of $Cu^{2+} \rightarrow Cu^{+}$]

Protein

Sample + $Cu^{2+} + OH^- \rightleftharpoons$ (Purple Colour)

* If protein is absent, then soln remains blue due $CuSO_4 \cdot 5H_2O$

Disadvantages =>

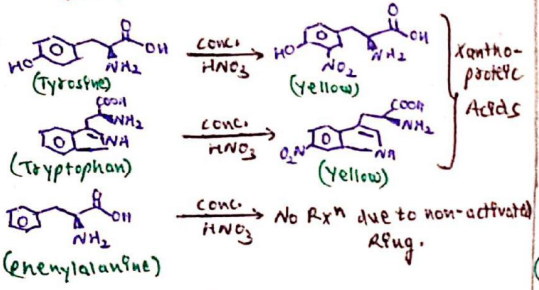
(i) Low sensitivity

(ii) Can't be used for proteins that precipitate by forming $(NH_4)_2SO_4$.

3) Xanthoproteic Test :->

* Used to detect aromatic amino acid with activated rings.

Reagent: Conc. HNO_3



* 'Dark yellow colour' indicates +ve test.

* On addn of alkali, xanthoproteic acids give orange colour.

Xanthoproteic acids (Yellow) $\xrightarrow{OH^-}$ Orange colour

* On dropping conc. HNO_3,

On skin => Lfgh yellow, easily removed

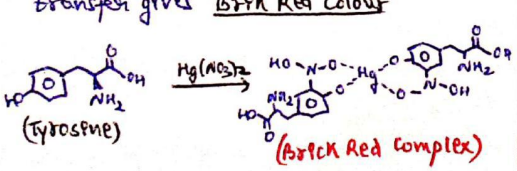
On nails => Dark yellow, not easily removed due to Keratin.

4) Millon's Test :->

* Used to detect Tyrosine (Phenolic Group)

Reagent: HNO_3 / $HgNO_3$ or $Hg(NO_3)_2$

* Nitration & complex formation after proton transfer gives 'Brick Red Colour'



* Millon's test can detect any compound with phenolic segment, thus prior to it; we perform Biuret or Ninhydrin test to confirm Amino acid.

Liebig's Method :

* For estimation of C and H in organic compound

Organic Compound containing C & H

- React with C (CuO): $C + CuO \rightarrow CO_2 + Cu$ (Lime water) $\downarrow Ca(OH)_2$ → Milky $CaCO_3$ → (Excess CO_2, Δ) → $Ca(HCO_3)_2$ (Milkyness disappears); Milkyness Reappears
- React with H (CuO): $H + CuO \rightarrow H_2O + Cu$ ($\downarrow$ anhyd. $CuSO_4$) → $CuSO_4 \cdot 5H_2O$ (Blue Vitrol)

* It is actually $[Cu(H_2O)_4]SO_4 \cdot H_2O$ that contains
- ✓ Covalent Bond
- ✓ Coordinate Bond
- ✓ Ionic Bond
- ✓ H-Bond.

NOTE: Coordination Number of $CuSO_4 \cdot 5H_2O$ is 4 and not 6

Lassaigne's Test :

-> Test for Detection of (P, S, N, X)

-> In first step of this test, sample is reacted with Na and the product formed is called 'Sodium Extract / Lassaigne Extract'

(I) For Nitrogen =>

(Sodium Extract)

$C + N + Na \xrightarrow{\Delta} NaCN$

$CN^- + Fe^{2+} \rightarrow [Fe(CN)_6]^{4-}$

$[Fe(CN)_6]^{4-} \xrightarrow[x \cdot H_2O]{Fe^{3+}} Fe_4^{+3}[Fe(CN)_6]^{+2}_3 \cdot x H_2O$ (Prussian Blue)

(II) For Halogens (X = Cl, Br, I) =>

$X + Na \xrightarrow{\Delta} NaX$

$X^0 + AgNO_3 \rightarrow AgX$

- AgCl (White ppt) → excess NH_4OH → $[Ag(NH_3)_2]^+Cl^-$ (soluble)
- AgBr (Pale yellow ppt) → excess NH_4OH → Sparingly soluble
- AgI (Dark yellow ppt) → excess NH_4OH → Insoluble

(III) For Sulphur (S) =>

$S + Na \xrightarrow{\Delta} Na_2S$

a) $S^{2-} + Pb^{2+} \rightarrow PbS\downarrow$ (Black ppt)

b) Sodium Nitroprusside Test =>

$Na_2S + Na_1[Fe^{+2}(CN)_5 N O^{+1}] \rightarrow Na_4[Fe^{+2}(CN)_5 NOS]$

(Sodium Nitroprusside) (Violet Coloured)

(IV) For Phosphorus (P) =>

$P \xrightarrow{Na_2O_2} PO_4^{3-} \xrightarrow{HNO_3} H_3PO_4$

$(H_3PO_4 + (NH_4)_2MoO_4 + HNO_3)$

$\downarrow$

$(NH_4)_3PO_4 \cdot 12MoO_3 \downarrow + NH_4NO_3$

(Canary Yellow ppt) (Ammonium phosphomolybdate)

* In case of the organic compound contains all (S, C, N) at the same time, then.

$S + C + N \xrightarrow[\Delta]{Na} NaSCN$

- excess Na → $NaCN + Na_2S$ (Both N & S can be identified)
- Fe^{3+} → $[Fe(SCN)]^{2+}$ or $[Fe(SCN)_3]$ (Blood Red complex)

Quantitative Analysis :

1) Liebig's Method for (% C) & (% H) :->

Let, W_1 gm → CO_2; W_2 gm → H_2O; W gm → organic sample

$C + CuO \rightarrow CO_2$

$H + CuO \rightarrow H_2O$

So,

$$\%C = \frac{12}{44} \times \frac{W_1}{W} \times 100$$

$$\%H = \frac{2}{18} \times \frac{W_2}{W} \times 100$$

2) Duma's Method for (% N) :->

$C_xH_yN_z + \left(2x + \frac{y}{2}\right)CuO \rightarrow x \cdot CO_2 + \frac{y}{2}H_2 + \frac{z}{2}N_2 + \left(2x + \frac{y}{2}\right)Cu$

oxides of N $\xrightarrow{CuO}$ N_2

Let, P = Pressure of moist gas

P_1 = Aqueous Tension (P_{H_2O})

V_1 = Vol. of N_2 collected at T_1 Kelvin.

So we can write

$$\frac{P - P_1}{T_1} = \frac{P_2 V_2}{T_2} \xrightarrow{@STP} V_2 = \frac{(P - P_1)V_1}{T_1} \times \frac{273}{760}$$

And,

$$\%N = \frac{28}{22400} \times \frac{V_2\,(ml)}{W} \times 100$$

W → Mass of organic sample

3) Kjeldahl's Method for (% N) :->

Organic Compound $\xrightarrow[CuSO_4]{H_2SO_4}$ $(NH_4)_2SO_4$ $\xrightarrow{NaOH}$ NH_3 (aq.)

NH_3 (aq.) $\xrightarrow{H_2SO_4}$ $(NH_4)_2SO_4$

So,

$$\%N = \frac{1.4}{W} \times N \times V\,(ml)$$

where, W → Mass of Organic Compound

N → Normality of acid used (Molarity × nf)

V → Vol. of acid used (in ml)

* This method cannot be used for compounds containing

1) Nitro Group ($-NO_2$)
2) Azo Group ($-N=N-$)
3) Nitrogen in Ring (Pyridine)

4) Carius Method for (% Cl / Br / I) :->

Organic Compound $\xrightarrow{AgNO_3}$ AgX

Now,

$$\%Cl = \frac{35.5}{108 + 35.5} \times \frac{W_1}{N} \times 100$$

$$\%Br = \frac{80}{108 + 80} \times \frac{W_1}{W} \times 100$$

$$\%I = \frac{127}{108 + 127} \times \frac{W_1}{W} \times 100$$

where, W_1 → Weight of AgX formed

W → Weight of Organic compound.

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5) Carius Method for (% S) :→

Organic compound $\xrightarrow[\text{Fuming } HNO_3]{Na_2O_2 \text{ or}} H_2SO_4 \xrightarrow{BaCl_2} BaSO_4$

$$\%S = \frac{32}{233} \times \frac{W_1}{W} \times 100$$

where, W_1 → Weight of $BaSO_4$ formed
W → Weight of Organic compound

6) Carius Method for (% P) =>

i) Organic compound $\xrightarrow[aq.\ NH_3]{MgSO_4} Mg(NH_4)PO_4 \xrightarrow{\Delta} Mg_2P_2O_7$

$$\%P = \frac{62}{222} \times \frac{W_1}{W} \times 100$$

where, W_1 → Weight of $Mg_2P_2O_7$ formed
W → Weight of Organic compound

ii) Organic compound $\xrightarrow{HNO_3} H_3PO_4 \xrightarrow{(NH_4)_2MoO_4} (NH_4)_3PO_4 \cdot 12MoO_3$

$$\%P = \frac{31}{1877} \times \frac{W_1}{W} \times 100$$

where, W_1 → Weight of $(NH_4)_3PO_4 \cdot 12MoO_3$
W → Weight of Organic Compound

Purification of Organic Compounds :

Methods to Purify Organic Compounds =>

1) Sublimation
2) Crystallisation
3) Distillation
4) Chromatography
5) Differential extraction.

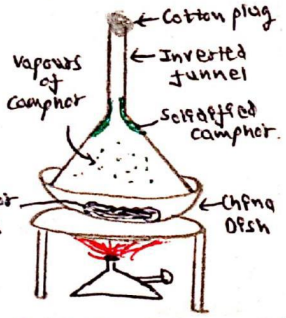
Sublimination :

→ On heating, some solid substances change from solid to vapour state w/o passing through liq. state.

→ Purification techique based on above principle is known as 'Sublimination'

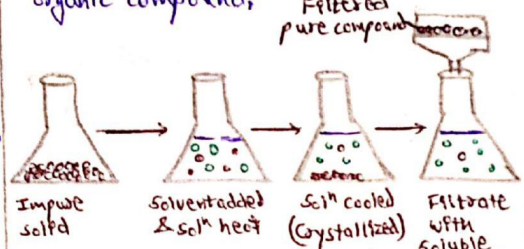
→ Used to separate,
Sublimable => Compounds
Non-Sublimable => Impurities.

Eg: (Sublimable)
i) Naphthalene and Anthracene
ii) Camphor from mix. of salts.



Crystallization :

* Used for purification of 'solid organic compound'
* Based on difference of 'solubilities' of the compound and impurities in a suitable solvent.
* Impure compound is first dissolved in a solvent in which it is sparingly soluble at room temp. but appreciably soluble at high temp.
* So, we heat this sparingly dissolved soln until we get nearly saturated soln.
* As soon as we get such soln, we quickly cool down the soln which causes crystallization of pure compound.
* After filtering crystals out, the filtrate contains impurities and small quantity of organic compound.



Important Points :→

1) Crystallisation can be satisfactorily carried out in a mix. of solvents such that it contain
   - Solvent in which compound is highly soluble
   - As well as solvent in which compound is very less soluble.
2) Impurities that impart colour are removed by 'adsorption over activated charcoal'
3) If compound and impurities have comparable solubilities then repeated crystallization (Recrystallization) is needed.

Distillation :

* Used to separate
  (i) Volatile liquids from non-volatile impurity.
  (ii) Liquids having sufficient diff. in their BP.

(I) Simple Distillation :→

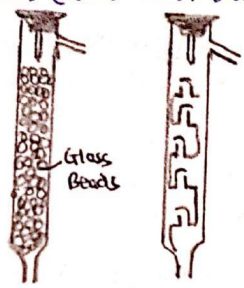
→ Liquids having different BP vaporise at different temp.

→ So, when such mix is heated, component with low BP vaporises first whose vapours are then condensed by cooling & collected separately.

→ Used for separation of
Chloroform (BP = 334K) and Aniline (BP = 457K)

(II) Fractional Distillation :→

→ If difference in BP of two liquids is very less (≤ 20 units), then we use this technique



* For fractional distillation, two types of fractionating columns can be used
* The vaporised soln collides with beads/columns and the component with high BP (easy to condense) gets condensed again & again.
* The condensation of component having high BP occurs throughout the column as vapours of soln keeps moving upward.
* As a result, at the top, only vapours of liquid with low BP are able to reach which are further condensed & collected separately.

Imp (Used to separate Crude Oil in Petroleum Industry)

(III) Distillation Under Reduced Pressure :→

→ Used to purify liquids that have high BP and which decompose at or below their own BP.

→ BP is the temp at which
$(\text{Vapour Pressure})_{liq}$ = External Pressure

→ The ext. pressure is reduced with the help of a 'Water pump/Vacuum pump'.

→ Hence, the liq. vaporises at a lower temp. w/o decomposing.

Glycerol separated from spent lye in soap industry by this technique

(IV) Steam Distillation :→

→ Used to separate substances which are steam volatile and are immiscible with water.

→ Steam from a steam generator is passed through a heated flask containing the liq. to be distilled. It does two things

① The water vap. ↑ses overall vap. pressure of heated flask due to which external vap. pressure becomes equal to Pext. at less temp only

② Water vap. dissolves the water soluble component in them from vapours of soln to be distilled.

→ The vapours are then condensed and compound is separated from H_2O using 'Separating Funnel'

→ Separation of 'Aniline' is carried out by steam distillation.

#Differential Extraction :

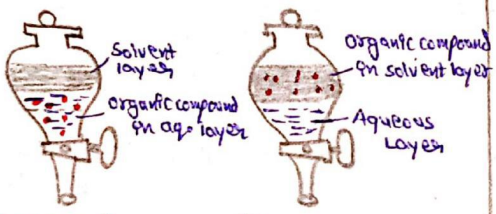
→ Used to separate organic compound from aqueous medium.

→ We add an organic solvent in the separating funnel along with aqueous soln, which is

① Immiscible with water

② organic compound is more soluble in this solvent than ~~in~~ water.

→ Then we shake this mixture, and due to density difference, two layers are formed in the separating funnel.



→ Later solvent is removed by Distillation or evaporation to get back organic compound

→ If compound is very less soluble in solvent, then very large amount of solvent is required and continuous repeated extraction is carried out.

#Chromatography :

→ In this technique, the mixture of substance is applied onto a 'stationary phase' which may be solid or liquid

→ A pure solvent, mixture of solvents or gas is allowed to move slowly over stationary phase.

→ The components of the mixture get gradually separated from one another. The moving phase is called 'mobile phase'

→ Based on principle used, chromatography is classified into two categories.

Adsorption Chromatography | Partition Chromatography

#Adsorption Chromatography :

* Based on difference in degree of adsorption of compounds on same adsorbent (stationary phase)

* Commonly used adsorbent are
1) Silica Gel 2) Alumina (Al_2O_3)

* On the basis of differential adsorption, two main chromatographic techniques are used.

Column Chromatography | Thin Layer Chromatography

(I) Column Chromatography :→

→ First, the mixture to be separated is adsorbed on our adsorbent.

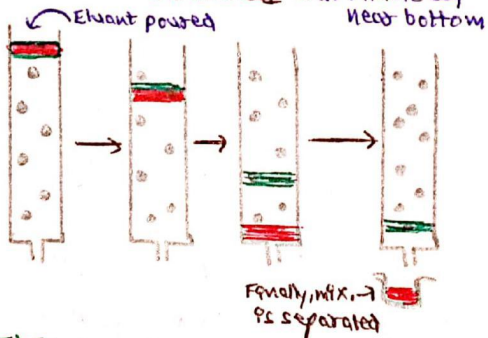
→ Then adsorbed mix. is placed on the top of the 'adsorbent column' packed in glass tube.

→ An appropriate eluant (liq. or mixture of liq.) is allowed to flow down the column.

→ Depending upon degree of adsorption,

Easily adsorbed substance ⇒ Near the top

Other substances → at middle or near bottom



(II) Thin Layer Chromatography :→

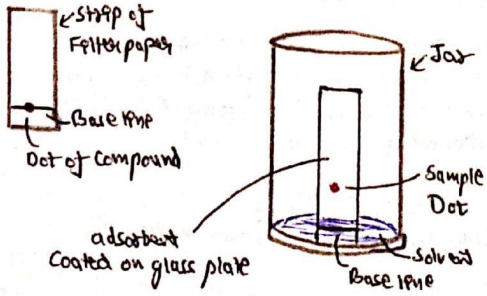
* First, we take a strip of 'Filter paper' and draw our base line with pen/pencil.

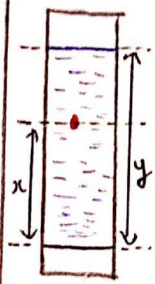
* On this base line, we make a dot of our compound (coloured or non coloured).

* Then, this paper is fixed on an adsorbent coated glass plate and placed in a jar.

* An appropriate solvent is then poured in the jar (acts as mobile phase), which gradually takes the 'dot' from base line to some height.

* The 'dot' stops moving once it gets adsorbed on the adsorbent.





* Once the Dot stops moving, we take out the strip and calculate R_f.

R_f = Distance moved by dot of substance from base (x) / Distance moved by the solvent from base line (y)

Non-Coloured Compound ⇒

→ If compound is coloured, its movement is easy to observe.

→ But if it is non-coloured then

1) Some compound give fluoresce in UV light

2) Some adsorb Iodine to give brown spots. This glass plate placed in jar containing Iodine crystals.

3) Sometime, some reagent can also be sprayed.

(Ninhydrin sprayed for Amino acids)

#Partition/Paper Chromatography :

→ Based on continuous differential partitioning of components of a mixture b/w stationary and mobile phase.

→ Special 'Chromatography paper' is used here, which has water trapped in it (water serves as stationary phase).

→ A strip of chromatography paper 'spotted' at base with sample soln is placed in suitable solvent.

→ Due to 'Capillary Action', solvent moves up and selectively differentiates components (solvent behaves as mobile phase).

→ Strip obtained after differential adsorption of all components with different coloured lines is called 'Chromatogram'

